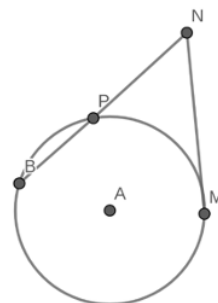


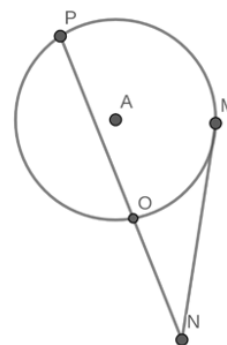
Geometry
Unit 12
Lesson 3 Practice

Name _____
Date _____
Period _____

1. Circle A is shown with points M, B and P on the circle. Secant $\overline{BN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PB = 5$ and $PN = 4$. What is the length of $\overline{MN}$?

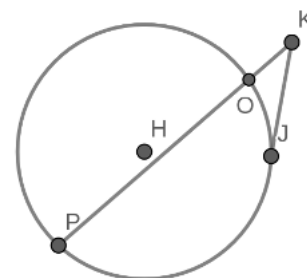


2. Circle A is shown with points M, O and P on the circle. Secant $\overline{PN}$ and tangent $\overline{MN}$ intersect at point N outside the circle. $PO = (3x + 3)$, $ON = 10$ and $MN = 20$. Find the value of x .



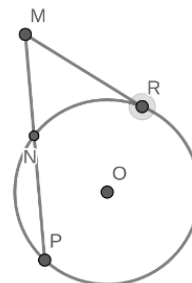
3. What is the length of $\overline{PN}$?

4. Circle H is shown with points J, O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 112$, $OK = 14$ and $KJ = x$. Find the value of x .



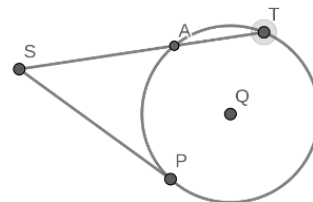
5. Using the same image as question 4, circle H is shown with points J, O and P on the circle. Secant $\overline{KP}$ and tangent $\overline{KJ}$ intersect at point K outside the circle. $PO = 16 + x$, $OK = 2$ and $KJ = 8$. Find the length of $\overline{KP}$.

6. Circle O is shown with points N, R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = 20$, $MN = 16$ and $MR = 4x$. What is the value of x ?



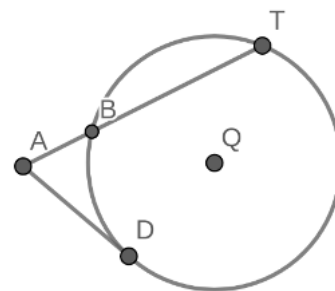
7. Using the same image as question 6, Circle O is shown with points N, R and P on the circle. Secant $\overline{MP}$ and tangent $\overline{MR}$ intersect at point M outside the circle. $PN = x + 10$, $MN = 15$ and $MR = 30$. What is the value of x ?

8. Circle Q is shown with points A, T and P on the circle. Secant $\overline{ST}$ and tangent $\overline{SP}$ intersect at point S outside the circle. $SP = 4x$, $AT = 18$ and $AS = 32$. What is the value of x ?



9. Daniella and Taylor are working on finding the length of $\overline{AD}$. Circle Q is shown with points B, T and D on the circle. Secant $\overline{AT}$ and tangent $\overline{AD}$ intersect at point A outside the circle. $BT = 12$, $AB = 10$ and $AD = 2x$. Their partial work solving for x is shown below. Find the value of x for each of their work. Round to the nearest hundredth. Whose work do you agree with?

Daniella's work	Taylor's work
$(2x)^2 = 10(10 + 12)$ $4x^2 = 10(22)$ $x^2 = 55$	$(2x)^2 = 12(10 + 12)$ $4x^2 = 12(22)$ $x^2 = 132$



10. Using the image in Question 9, what is the length of $\overline{AD}$?